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GRAU DE MATEMÀTIQUES

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GRAPH NEURAL NETWORKS FOR AUTONOMOUS THEOREM PROVING

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Abstract

The main goal of this work is to prove the existence of.....

Notation: We will work over an algebraically closed field K of characteristic zero.

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Chapter 1

Generalized Algebraic Theories

1.1 Definition

We assume we have a set V of countably many distinct variables. Let W be an **alphabet**, and its elements be **symbols**.

Definition 1.1. *The set of **expressions** is defined inductively such that they are finite sequences of elements of $W \cup V \cup \{(\} \cup \{)\} \cup \{,\}$ by the clauses*

1. *if $x \in V$, then x is an expression;*
2. *if $L \in W$, then L is an expression;*
3. *if $L \in W$ and $e_1, \ldots, e_n$ are expressions, then $L(e_1, \ldots, e_n)$ is an expression.*

Definition 1.2. *A **premise** is defined to be a sequence of $V \times$ set of expressions, the empty sequence is allowed and is called the **empty premise**. A premise $((x_1, \Delta_1), \ldots, (x_n, \Delta_n))$ is written as $x_1 : \Delta_1, \ldots, x_n : \Delta_n$.*

Definition 1.3. *A **conclusion** is one of the following:*

1. *T**-conclusion** consists a single expression Δ and is written as "Δ is a type";*
2. *$\in$**-conclusion** consists of pair of expressions (t, Δ) and is written as "$t : \Delta$";*
3. *$T =$**conclusion** consists of a pair of expressions (Δ_1, Δ_2) and is written as "$\Delta_1 = \Delta_2$";*
4. *$\in =$**conclusion** consists of a triple of expressions (t_1, t_2, Δ) and is written as "$t_1 = t_2 \in \Delta$".*

Definition 1.4. *A **rule** is determined by a premise P and a conclusion C, and is written as*

$$\frac{P}{C}$$

If $\Delta, t_1, \ldots, t_n$ expressions and $x_1, \ldots, x_n$, then the expression $\Delta[t_1|x_1, \ldots, t_n|x_n]$ is the result of simultaneous replacement of $x_1, \ldots, x_n$ in Δ by $t_1, \ldots, t_n$.

Definition 1.5. *A **pretheory** consists of*

1. *a set S, of sort symbols;*
2. *a set Z, of operator symbols;*
3. *for each sort symbol A, an associated rule of the alphabet $S \cup Z$, called **introductory rule** for A, of the form*

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{A(x_1, \ldots, x_n) \text{ is a type}}$$

   for some $n \geq 0$;

4. *for each operator symbol F, an associated rule of the alphabet $S \cup Z$, called **introductory rule** for F, of the form*

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{F(x_1, \ldots, x_n) \in \Delta}$$

   for each $n \geq 0$;

5. *a set of axioms, each axiom is either a $T =$rule or a $\in$-rule of the alphabet $S \cup Z$.*

Definition 1.6. *If U is a pretheory, then*

1. *a **context** is a premise $x_1 : \Delta_1, \ldots, x_n : \Delta_n$ such that the rule*

$$\frac{x_1 : \Delta_1, \ldots, x_{n-1} : \Delta_{n-1}}{\Delta_n \text{is a type}}$$

   is a derived rule such that x_n is distinct from all of $x_1, \ldots, x_{n-1}$.

2. *TODO*

Definition 1.7. *The set of **derived rules** of U is the set of rules derivable by the following principles of derivation*

LI1 *From* $\dfrac{P}{\Delta \text{ is a type}}$ *derive* $\dfrac{P}{\Delta = \Delta}$.

LI2 *From* $\dfrac{P}{t : \Delta}$ *derive* $\dfrac{P}{t = t : \Delta}$.

LI3 *From* $\dfrac{P}{\Delta_1 = \Delta_2}$ *derive* $\dfrac{P}{\Delta_2 = \Delta_1}$.

LI4 *From* $\dfrac{P}{t_1 = t_2 : \Delta}$ *derive* $\dfrac{P}{t_2 = t_1 : \Delta}$.

LI5 *From* $\dfrac{P}{\Delta_1 = \Delta_2}$ *and* $\dfrac{P}{\Delta_2 = \Delta_3}$ *derive* $\dfrac{P}{\Delta_1 = \Delta_3}$.

LI6 *From* $\dfrac{P}{t_1 = t_2 : \Delta}$ *and* $\dfrac{P}{t_2 = t_3 : \Delta}$ *derive* $\dfrac{P}{t_1 = t_3 : \Delta}$.

LI7 *From* $\dfrac{P}{t_1 = t_2 : \Delta_1}$ *and* $\dfrac{P}{\Delta_1 = \Delta_2}$ *derive* $\dfrac{P}{t_1 = t_2 : \Delta_2}$.

T1 *From* $\dfrac{P}{\Delta_1 = \Delta_2}$ *and* $\dfrac{P}{t : \Delta_1}$ *derive* $\dfrac{P}{t : \Delta_2}$.

CF1 *For* $n \geq 0, 1 \leq i \leq n+1$*:*
From $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta_{n+1} \textit{is a type}}$ *derive* $\dfrac{x_1 : \Delta_1, \ldots, x_{n+1} : \Delta_{n+1}}{x_i : \Delta_i}$.
providing that x_{n+1} *is a variable distinct from all of* $x_1, \ldots, x_n$.

CF2(a) *For every sort symbol* A *with well-formed introductory rule*
$\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{A(x_1, \ldots, x_n) \textit{is a type}}$*, for every context* P*,*
from $\dfrac{P}{t_1 : \Delta_1}, \dfrac{P}{t_2 : \Delta[t_1|x_1]}, \ldots, \dfrac{P}{t_n : \Delta_n[t_1|x_1, \ldots, t_{n-1}|x_{n-1}]}$
derive $\dfrac{P}{A(t_1, \ldots, t_n) \textit{is a type}}$.

CF2(b) *For every operator symbol* F *with well-formed introductory rule*
$\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{F(x_1, \ldots, x_n) : \Delta}$*, for every context* P*,*
from $\dfrac{P}{t_1 : \Delta_1}, \dfrac{P}{t_2 : \Delta_2[t_1|x_1]}, \ldots, \dfrac{P}{t_n : \Delta_n[t_1|x_1, \ldots, t_{n-1}|x_{n-1}]}$
derive $\dfrac{P}{F(t_1, \ldots, t_n) : \Delta[t_1|x_1, \ldots, t_n|x_n]}$.

SI1 *If* Q *is a context, then from* $\dfrac{y_1 : \Omega_1, \ldots, y_n : \Omega_n}{\Omega = \Omega'}$ *and from*
$\dfrac{Q}{s_1 = s_2 : \Omega_1}, \dfrac{Q}{s_2 = s_2' : \Omega_2[s_1|y_1]}, \ldots, \dfrac{Q}{s_m = s_m' : \Omega_m[s_1|y_1, \ldots, s_{m-1}|y_{m-1}]}$
derive $\dfrac{Q}{\Omega[s_1|y_1, \ldots, s_m|y_m]} = \Omega'[s_1'|y_1, \ldots, s_m'|y_m]$.

SI2 *If* Q *is a context, then from* $\dfrac{y_1 : \Omega_1, \ldots, y_m : \Omega_m}{s = s' : \Omega}$ *and from*
$\dfrac{Q}{s_1 = s_1' : \Omega_1}, \dfrac{Q}{s_2 = s_2' : \Omega_2[s_1|y_1]}, \ldots, \dfrac{Q}{s_m = s_m' : \Omega_m[s_1|y_1, \ldots, s_{m-1}|y_{m-1}]}$

derive $\dfrac{Q}{s[s_1|y_1,\ldots,s_m|y_m] = s'[s'_1|y_1,\ldots,s'_m|y_m] : \Omega[s_1|y_1,\ldots,s_m|y_m]}$.

A1 *If* $\dfrac{x_1 : \Delta_1,\ldots,x_n : \Delta_n}{\Delta = \Delta'}$ *is an axiom then from* $\dfrac{x_1 : \Delta_1,\ldots,x_n : \Delta_n}{\Delta \textit{is a type}}$ *and* $\dfrac{x_1 : \Delta_1,\ldots,x_n : \Delta_n}{\Delta'}$.

A2 *If* $\dfrac{x_1 : \Delta_1,\ldots,x_n : \Delta_n}{t = t' : \Delta}$ *is an axiom, then from* $\dfrac{x_1 : \Delta_1,\ldots,x_n : \Delta_n}{t : \Delta}$ *and* $\dfrac{x_1 : \Delta_1,\ldots,x_n : \Delta_n}{t' : \Delta}$ *derive* $\dfrac{x_1 : \Delta_1,\ldots,x_n : \Delta_n}{t = t' : \Delta}$

Chapter 2

títol del capítol 2

Bibliography